Decent Coaching Classes

VIII SB

Academic Conquest - III

Answer Key — Chapter 5: Inside the Atom

Q.1 Fill in the blanks — 1 mark each

a. indivisible, indestructible

b. J.J. Thomson

c. 1911

d. $2n^2$ (2 times n squared)

e. Isotopes

f. Graphite or heavy water

Q.2 Answer in one sentence — 1 mark each

a. An atom is the smallest particle of an element which retains its chemical identity in all physical and chemical changes.

b. J.J. Thomson demonstrated that the negatively charged particle (electron) in an atom has a mass 1800 times less than that of a hydrogen atom.

c. The electrons present in the outermost shell (valence shell) of an atom are called valence electrons.

d. Any two: Uranium-235 is used for nuclear fission and production of electricity; Cobalt-60 is used in the medical treatment of cancer; Iodine-131 is used in the treatment of goiter; Sodium-24 is used to detect leakage in underground pipes; radioactive isotopes are used for food preservation; C-14 is used to determine the age of archaeological objects.

e. The atomic number of an element is the number of protons present in the nucleus of its atom, denoted by 'Z'.

f. Proton and Neutron.

Q.3 Match the pairs — 4 marks (1 mark each)

a. → ii. Positively charged particle

b. → iv. Negatively charged particle

c. → i. Electrically neutral particle

d. → iii. Same atomic number, different mass number

Q.4 Give scientific reasons — 2 marks each (any 3 of 4)

a. Almost the entire mass of an atom is concentrated in its nucleus because the nucleus contains the protons and neutrons (nucleons), each of mass approximately 1 u, while the electrons revolving in the extranuclear part have negligible mass — about 1800 times less than that of a hydrogen atom. Hence protons and neutrons together account for almost all of the atom's mass.

b. An atom is electrically neutral because the number of electrons (negatively charged) present in the extranuclear part is exactly equal to the number of protons (positively charged) present in the nucleus. Since the total negative charge on the electrons balances the total positive charge on the nucleus, the atom as a whole carries no net charge.

c. The atomic mass number of an element is a whole number because it represents the total count of protons and neutrons (nucleons) in the nucleus, and the number of particles can only be expressed as a whole number, never as a fraction.

d. According to an established law of physics, a charged body moving in a circular orbit continuously loses energy. In Rutherford's model, electrons revolve around the nucleus in such circular paths, so they should continuously lose energy and eventually spiral into the nucleus, making the atom unstable. Since real atoms (other than radioactive ones) are stable, Rutherford's model was considered unstable/inadequate — this shortcoming was later removed by Bohr's atomic model.

Q.5 Distinguish between — 2 marks each (any 2 of 3)

a. Thomson's model: positive charge is spread throughout the atom with negatively charged electrons embedded in it (like plums in a pudding); it has no nucleus. Rutherford's model: positive charge and almost all the mass are concentrated in a small central nucleus, with electrons revolving around it in mostly empty space.

b. Proton: a positively charged subatomic particle found in the nucleus, denoted 'p', carries a charge of +1e, mass about 1u. Neutron: an electrically neutral subatomic particle found in the nucleus, denoted 'n', carries no charge, mass about 1u (approximately equal to a proton's mass).

c. Nucleus: the small, dense, positively charged central part of the atom containing protons and neutrons, holding almost the entire mass of the atom. Extranuclear part: the region outside the nucleus containing electrons revolving around it and the empty space between the nucleus and the electrons.

Q.6 Answer the following — 3 marks each (any 2 of 3)

a. Bohr's stable orbit atomic model (1913) — postulates: (i) Electrons revolve around the nucleus in concentric circular orbits (shells) at certain fixed distances from the nucleus. (ii) The energy of an electron remains constant as long as it stays in a particular orbit/shell. (iii) When an electron jumps from an inner orbit to an outer orbit, it absorbs energy equal to the difference between the two energy levels; when it jumps from an outer orbit to an inner orbit, it emits energy equal to that difference. This model explained the stability of the atom, which Rutherford's model could not.

b. The valency of an element is related to the number of valence electrons in its atom: if the number of valence electrons is four or less than four, the valency of the element is equal to that number of valence electrons. If the number of valence electrons is four or more than four, the valency is equal to $(8 - x)$, where x is the number of valence electrons — i.e., the number of electrons needed to complete the octet.

c. Atomic number (Z) = 12, Mass number (A) = 24. Number of protons = Z = 12. Number of electrons = number of protons (atom is neutral) = 12. Number of neutrons = $A - Z = 24 - 12 = 12$. Electronic configuration: distribute 12 electrons into shells K(2), L(8), M(18 max) → K = 2, L = 8, M = 2. So the electronic configuration is 2, 8, 2 (this corresponds to magnesium, Mg).

Total: 30 Marks